Finding disease-proximal genes

Trans-regulatory gene mapping prioritizes disease drivers in asthma

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Journal Club | DANDELION

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The strongest association may be an upstream regulator

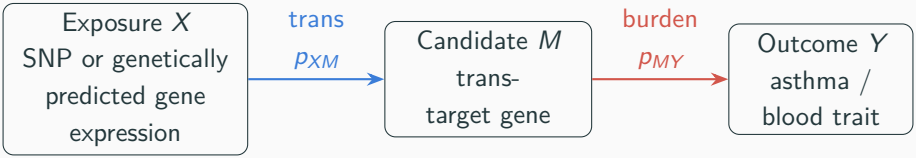
Question: Which genes sit closest to the mechanisms that produce disease?

- GWAS can identify many risk loci, but a locus may regulate several downstream genes rather than execute disease biology itself.
- Rare-variant burden tests directly implicate coding genes, but modest sample sizes and rare alleles limit power.
- The authors call candidate downstream mediators disease-proximal genes (DPGs).

Contribution
Combine trans-regulation with rare-variant burden evidence, then experimentally test the resulting candidates in asthma.

Introduction; Fig. 1A–B. DPG is a proposed functional category, not an established causal label.

Two evidence channels meet at the same candidate gene



Output: prioritized mediator genes and exposure-to-target networks.

Crucial distinction: the burden statistic tests coding variation in M; it is not a direct estimate of an expression-mediated causal effect.

Fig. 1A–B; STAR Methods, DANDELION method. Diagram redrawn for teaching.

What goes into DANDELION?

Input	Gene-based implementation	SNP-based implementation
Trans evidence	GBAT predicts each exposure gene from cis SNPs using leave-one-out cvBLUP; tests targets on other chromosomes	Summary statistics of SNP-to-gene trans-eQTLs
Trait evidence	Gene-level WES burden p values for candidate target genes	
Integration	DACT composite-null test, calibration, per-exposure q values	

Large consortia make SNP summary statistics attractive; gene-based analyses need individual-level genotype and expression data.

STAR Methods, DANDELION method; Discussion. GBAT is a component method, not a new experiment here.

Why a composite null is needed

Let γ represent trans regulation and β the candidate's association with the trait.

Case	$X \to M$	$M \to Y$	Interpretation
1	$\gamma = 0$	$\beta \neq 0$	Trait gene; no measured trans link
2	$\gamma \neq 0$	$\beta = 0$	Regulated target; no trait evidence
3	$\gamma = 0$	$\beta = 0$	Neither edge supported
4	$\gamma \neq 0$	$\beta \neq 0$	Putative pathway

$$H_0 : \beta\gamma = 0 \qquad \text{versus} \qquad H_1 : \beta\gamma \neq 0.$$

Rejecting one edge's null alone does not establish a two-edge pathway.

STAR Methods, DANDELION method; Fig. 1B. Symbols describe the model used to motivate the test.

DACT combines evidence and calibrates the null

For candidate j, form a mixture over the three null cases:

$$D_j = \widehat{\omega}_1 p_{XM,j} + \widehat{\omega}_2 p_{MY,j} + \widehat{\omega}_3 \{\max(p_{XM,j}, p_{MY,j})\}^2.$$

- Estimate null-case proportions across genome-wide candidate genes.
- Transform D_j to a normal score and use an empirical null's mean and SD to calibrate the p value.
- Convert calibrated p values to q values within each exposure; nominate targets on significant paths.

Interpretation
This is a prioritization test. It estimates neither the size nor the direction of a causal mediation effect.

STAR Methods, DANDELION method. The paper uses the DACT framework of Liu et al. (2022).

A concrete reading of a nominated pathway

PKN3-associated locus $\longrightarrow$ *SCD* $\longrightarrow$ asthma

- A trans association connects an asthma-associated locus near *PKN3* to *SCD* expression.
- *SCD* has suggestive coding-variant burden evidence ($p = 0.002$), below conventional gene-wide significance.
- Joint evidence nominates *SCD* for experimental follow-up, despite strong constraint ($pLI = 1$).

What remains unknown: whether this exact human locus changes asthma risk through *SCD*, in which cell type, and with what mediated effect size.

Results, asthma network; Limitations; Fig. 2A. This example illustrates a nomination, not validated locus mediation.

Blood traits provide an interpretable first benchmark

- 27 UK Biobank blood traits; trans-eQTLs from eQTLGen ($n = 31{,}684$).
- 14–154 DPGs per trait at 5% per-exposure FDR.
- Reference: 314 curated Mendelian blood-disorder genes.
- DPGs show 5.3–21.7-fold enrichment versus gene-length-matched random genes.

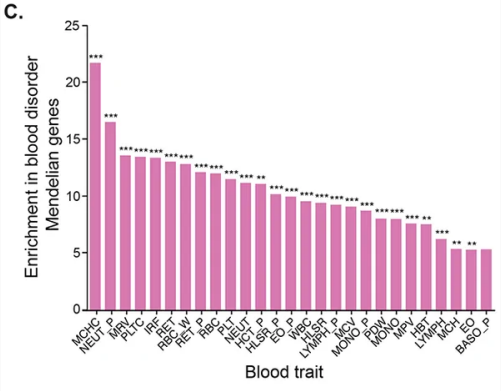
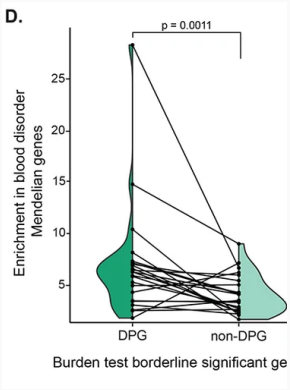


Fig. 1C; Table S1; Results, Mendelian enrichment. Fold enrichment is biological validation, not a calibrated precision estimate.

Does trans information add beyond a weaker burden threshold?



Control: compare DPGs with non-DPGs having similar burden-test significance.

DPGs retain stronger Mendelian-gene enrichment ($p = 0.0011$, paired comparison).

Inference: trans information changes which subthreshold genes are selected.

Caveat: Mendelian annotations are incomplete and blood traits have unusually strong prior biological knowledge.

Fig. 1D; Fig. S5 described in Results. Plot retained with original axes.

Asthma discovery integrates two whole-blood resources

	eQTLGen	DGN
Trans sample size	31,684	913
Implementation	SNP-based	Gene-based GBAT
Per-exposure FDR	5%	10%
DPGs nominated	13	16

Eight overlap $\Rightarrow$ 21 distinct candidates.

Burden data: 61,807 cases and 313,658 controls of European ancestry; rare predicted LoF + deleterious missense variants, MAF $< 1\%$.

Unlike the blood-trait analysis, asthma used genome-wide exposures to maximize discovery for experimental follow-up.

Results, asthma discovery; STAR Methods, Identify asthma DPGs; Fig. 2A; Table S4.

What is new, and what is replicated?

- Standard burden tests identified *FLG* and *IL*33; neither was nominated by DANDELION. DPG burden p values span 4×10^{-6} to 3×10^{-3}.
- Only three DPGs were nearest genes to significant UK Biobank asthma GWAS loci: *FAM*105*A*, *NRROS*, *ARHGAP*27.
- In independent primary $CD4^{+}$ T-cell perturb-seq, 16 DPGs were expressed; all 16 passed $p < 0.05/16$ after re-estimating trans effects.
- Removing high-degree trans regulators/targets retained all DPGs except *TAF*6*L*.

Qualification: perturb-seq replication changes the trans evidence; it is not an independent replication of the disease burden association.

Results, asthma discovery; Fig. S11 and hub-removal analysis described in full text; Table S4.

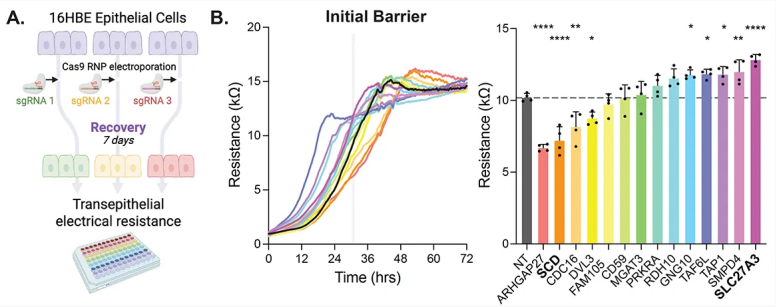
From whole blood to relevant cell phenotypes

Cell-context hypothesis: Human Lung Cell Atlas expression implicates immune cells and alveolar type 2 epithelium. Expression alone does not establish functional relevance.

System	Perturbation design	Readout and rationale
16HBE bronchial epithelial line	Arrayed CRISPR; expressed DPGs, distal genes and controls	Electrical resistance: formation and maintenance of epithelial barrier
Primary human Th2 cells	Pooled CRISPR; 132 genes; three donors	Guide enrichment in top 15% IL-13-high cells: type 2 inflammatory output

Fig. 2B–C; Fig. 3A,E; STAR Methods, CRISPR screens. Cell line is bronchial, not an alveolar type 2 model.

DPG knockdown often changes epithelial barrier function

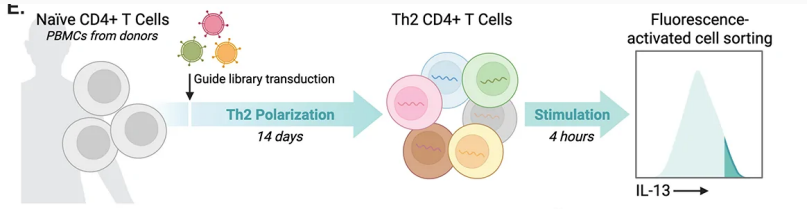


TEER changed for **10/14 DPGs (71.4%)**, versus 4/29 distal genes, 2/18 matched random genes, and 13/26 surviving fine-mapped GWAS genes.

Four additional GWAS-gene knockdowns were lethal: **17/30** affected barrier function or survival. These denominators answer different questions.

Fig. 3A–D; Fig. S14 described in Results. TEER: transepithelial electrical resistance. Changes may be beneficial or harmful.

A pooled Th2 screen tests cytokine output



- Compare guides in IL-13-high cells with all Th2 cells; *IL*13 and *GATA*3 are positive controls. Hit threshold: FDR < 0.25 plus a fold-change cutoff.
- Hits: 8/19 DPGs (42.1%), 7/30 GWAS genes (23.3%), 12/43 distal genes (27.9%), 9/38 nearest-neighbor controls (23.6%).
- *SLC*27*A*3 and *TAP*1 guides are strongly depleted from IL-13-high cells; *SMPD*4 guides are enriched.

Fig. 3E–G; Table S7; Results. Hit enrichment supports relevance but does not validate all 21 candidates.

Two candidates point to a shared lipid pathway

SLC27A3: fatty-acid transport protein and acyl-CoA synthetase; supplies palmitoyl-CoA.

SCD: desaturates palmitoyl-CoA to palmitoleic-CoA, potentially reducing substrate available for palmitoylation.

Prediction: their loss can have opposing effects. Screens support protective effects of *SLC27A3* loss and pathogenic effects of *SCD* loss.

Palmitoylation can alter protein localization, stability and signaling.

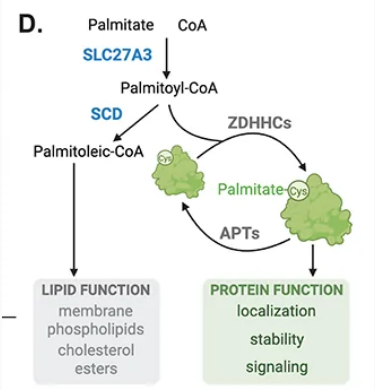
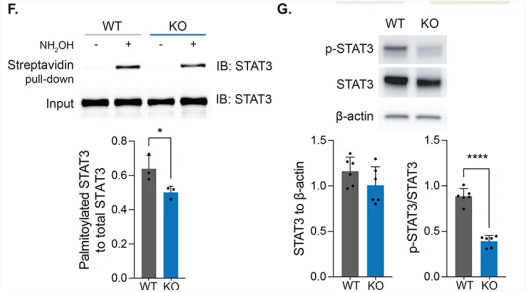


Fig. 4D; Results, lipid transport and palmitoylation. Pathway diagram reproduced from the paper.

Biochemistry connects SLC27A3 loss to STAT3 signaling



In 16HBE cells, *SLC27A3* loss reduces STAT3 palmitoylation and phosphorylated STAT3, with total STAT3 unchanged. Global probe labeling also decreases.

Mechanistic boundary: this supports a pathway connection; no rescue here proves that STAT3 palmitoylation alone explains the barrier or asthma phenotypes. *SCD* loss minimally changed total global palmitoylation.

Fig. 4E–G; acyl-biotin exchange, immunoblots and probe-labeling assays. Error bars: SD.

Slc27a3 loss attenuates allergen-induced remodeling

Model: germline knockout; intranasal *Alternaria* sensitization, two-week rest, challenge, day-20 assessment.

Reduced airway eosinophils ($p = 0.0220$) and reduced epithelial thickness, mucus and collagen.

Alt groups: 6 WT / 8 KO; PBS controls: 2 per genotype.

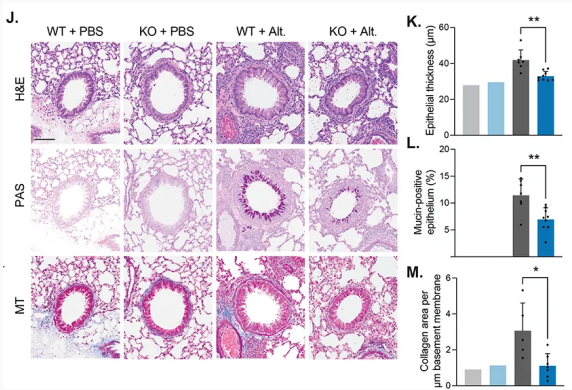
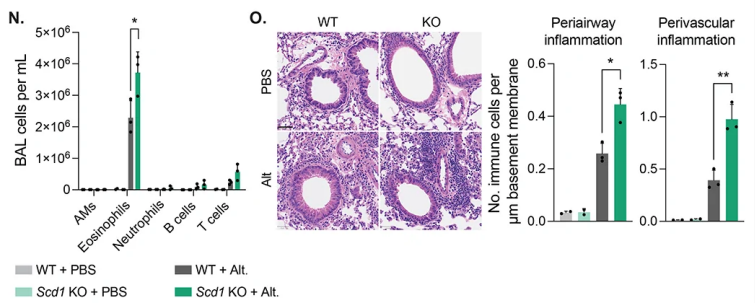


Fig. 4H–M; STAR Methods, Mice. Male Slc27a3 mice; histology panels retain original scale bar (100 micrometers).

Scd1 loss has the opposite inflammatory phenotype



Allergen-treated *Scd*1 knockouts have increased airway eosinophils and inflammation around airways and blood vessels.

Caution: three mice per allergen group; female *Scd*1 experiment versus male *Slc*27*a*3 experiment. These are separate genotype comparisons, not a direct cross-gene trial.

Fig. 4N–O; STAR Methods, Mice. PBS: 2 per genotype; histology scale bar 50 micrometers. Error bars: SD.

Evidence is strongest for prioritization plus functional relevance

Claim	Evidence supplied	What is still missing
Trans integration improves selection	Burden-matched Mendelian enrichment; novel candidates	Generalization outside these traits/resources
DPGs influence asthma biology	Barrier and Th2 screens; two mouse knockouts	Causal testing of each nominated human pathway
Palmitoylation is mechanistically relevant	Lipidomics, labeling, STAT3 assays, opposing phenotypes	Cell-specific rescue; separation from other lipid effects
Therapeutic opportunity	Human genetic support and protective knockout	Pharmacologic efficacy, timing, specificity and safety

Synthesis of Figs. 1–4; interpretation in right column is for journal-club discussion.

Statistical and biological limits affect how to use the list

1. **Multiplicity:** 5%/10% FDR is per exposure. The combined 21-gene list does not have that global guarantee; genome-wide asthma exposures broaden the search further.
2. **Context and dependence:** bulk trans-eQTLs can reflect cell composition; tissue mismatch, correlated exposures and model calibration affect interpretation.
3. **Coverage:** constrained genes may lack burden power; European-ancestry UK Biobank results do not establish portability; results depend on the chosen variant mask.

Functional enrichment is encouraging, but candidate selection and assay choice do not provide an unbiased estimate of genome-wide causal precision.

Discussion; Limitations; STAR Methods, simulations. Final sentence is our interpretation.

Discussion: what would make the mechanism convincing?

1. Can a human risk-allele perturbation alter the nominated DPG and phenotype, with DPG rescue reversing the effect?
2. Can cell-specific or adult-onset *Slc27a3* inhibition separate developmental effects from a therapeutically relevant response?
3. Which rescue distinguishes palmitoylation/STAT3 from arachidonic-acid uptake and other lipid-mediated mechanisms?

Take-home
DANDELION turns trans-regulatory connectivity and weak coding evidence into a useful experimental shortlist. Asthma experiments reveal actionable biology, while causal mediation and therapeutic benefit remain to be established.

Sources and reading scope

Verified citation

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Scope: full main text and STAR Methods read; figure crops from the PMC manuscript. Supplementary PDF and spreadsheet download endpoints were inaccessible during preparation. Supplementary results are cited only where described in the full text; underlying supplementary panels and tables were not independently inspected.

Slides distinguish statistical nominations, experiments and journal-club interpretation. Prepared November 1, 2026.